

Multi-Contrast Human Brain CEST MRI at 11.7 T: First In Vivo Demonstration

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Abstract

Purpose: To implement and evaluate Chemical Exchange Saturation Transfer (CEST) at ultra-high field on a whole-body 11.7 T scanner using parallel transmission (pTx).

Methods: Tailored pTx CEST saturation pulses were optimized to achieve a spatially uniform target B_1^+ while satisfying SAR and power constraints either defined by conservatively defined radio-frequency (RF) power limits or by a virtual observation point (VOP)-based framework. The resulting pulse performance at 11.7 T was assessed through numerical simulations and compared against corresponding designs at 7 T. The proposed saturation pulses were integrated into a two-dimensional multi-slice EPI sequence. Experimental validation was performed in vitro using metabolite-doped phantoms and in vivo in four healthy volunteers.

Results: The pTx pulse designs optimized under VOP-based SAR constraints at 11.7 T achieved comparable performance to 7 T designs despite higher intrinsic B_1^+ inhomogeneity. In vitro measurements at 11.7 T showed overall agreement with numerical simulations, with moderately increased variability attributable to the nonlinear sensitivity of MTR_{asym} to residual B_1^+ and B_0 inhomogeneities, as well as to unavoidable material heterogeneity in a gelatin-based phantom. In vivo experiments demonstrated acceptable saturation homogeneity ($nRMSE \leq 12.1\%$) in a slab of thickness 1 cm, with only minimal additional improvement achieved through B_1^+ post-processing correction. Across the cohort, robust gray-white matter contrast was consistently observed for multiple CEST contrasts.

Conclusion: VOP-based SAR-constrained pTx CEST saturation pulse design enabled robust and consistent contrasts at 11.7 T by effectively mitigating B_1^+ inhomogeneities while remaining within safety limits, facilitating translation to future in vivo clinical research applications.

Key words: chemical exchange saturation transfer, ultra-high field, parallel transmission, virtual observation points

1. Introduction

Chemical Exchange Saturation Transfer¹ (CEST) is a noninvasive MRI contrast mechanism that enables the indirect detection of low-concentration metabolites and mobile proteins by exploiting proton exchange with bulk water. Selective radio-frequency (RF) saturation of exchangeable solute protons at their resonance frequencies results in the transfer of saturation to bulk water protons, generating CEST contrast through changes in the water signal only. Prolonged saturation amplifies the resulting attenuation of the water signal, thereby enhancing sensitivity to molecular components that are difficult and time-consuming to image using conventional MR spectroscopic imaging² while eliminating the need for exogenous tracers, unlike positron emission tomography. CEST MRI has been used to probe a variety of molecules, such as amide protons in brain tumors^{3–5} or glutamate in the healthy brain at 7 T⁶ and is increasingly explored in clinical research in stroke⁷, epilepsy^{8,9}, and other neurologic and oncologic diseases^{10–13}. The launch in 2024 of the investigational whole-body 11.7 T Iseult MR scanner¹⁴ in France (NeuroSpin, CEA) offers a unique opportunity to further explore the capabilities of CEST MRI at extreme magnetic fields.

At high magnetic field strengths, both the longitudinal relaxation time of water and the spectral separation between resonances increase¹⁵, enhancing CEST sensitivity and enabling the detection of a broader range of CEST effects. In this context, glutamate-weighted CEST at 11.7 T is of interest, as its detectability at field strengths below 7 T remains debated due to its limited spectral dispersion^{6,16}. Similarly, hydroxyl-weighted CEST, with resonances as close as 1 ppm to water, may benefit substantially from the

increased spectral separation available at ultra-high field¹⁷, as previously demonstrated at 7 T¹⁸.

While ultra-high magnetic field strength provides clear benefits for CEST MRI, it also introduces significant technical challenges. In particular, spatial B_0 and B_1^+ inhomogeneities intensify, resulting in enhanced susceptibility effects, pronounced spatial variation in RF excitation, and increased RF energy deposition. Addressing these limitations has therefore become a major focus of recent methodological developments in ultra-high field CEST MRI.

The most efficient way to mitigate B_1^+ inhomogeneities is to use parallel transmission¹⁹ (pTx), which employs multiple independently-driven transmit channels. Dynamic RF shimming with pTx has been widely adopted in ultra-high field MRI, notably for slice-selective spoke excitations²⁰ and three-dimensional excitation schemes²¹ to reduce flip-angle nonuniformity. More recently, pTx has been extended to CEST MRI to improve saturation homogeneity at ultra-high field. The benefits of pTx for CEST imaging were demonstrated using the MIMOSA²² approach, which employs two complementary alternating static B_1^+ shim configurations to mitigate B_1^+ inhomogeneities. Further improvements were achieved using PUSH²³ saturation pulses, which optimize saturation homogeneity based on the measured B_1^+ distribution. While subject-specific pTx pulse optimization remains time-consuming, universal pulse²⁴ (UP) implementations of CEST saturation pulses have also been proposed in two-dimensional acquisitions at 7 T¹⁸, paving the way for simpler, faster and clinically feasible CEST protocols. The utility of such approaches was further demonstrated for whole-brain 3D CEST imaging at 7 T^{25,26}, using conservative vendor-defined specific absorption rate (SAR) limits for the pulse design. Delebarre et al.¹⁸ proposed the optimization of CEST saturation pulses using Virtual Observation Points²⁷ (VOP), enabling real-time SAR supervision and more efficient use of pTx capabilities at 7 T, resulting in substantial gains in CEST saturation uniformity of up to 4.7 fold.

The present work reports the first implementation and evaluation of CEST MRI on a human 11.7 T scanner, combining custom pulse sequence development, numerical

simulations, phantom validation experiments, and in vivo measurements. Tailored CEST pTx pulses and slice-specific three-spoke excitation pulses were designed to achieve optimal saturation under VOP-SAR constraints. The performance of the CEST pulses was evaluated through simulations and benchmarked against numerical results at 7 T. Several CEST contrasts were evaluated in realistic head phantoms doped with metabolites and in healthy volunteers. The efficiency of a B_1^+ post-processing correction²⁸ was also assessed on the acquired in vivo data.

2. Methods

2.1. Experimental set-up

All in vivo experiments were performed on the investigational whole-body 11.7 T Iseult MR scanner using an in-house head coil with 8 transmit (Tx) and 31 receive (Rx) channels²⁹. In vitro experiments were conducted on the same system, as well as on an investigational whole-body 7 T MR scanner equipped with a Nova head coil (Nova medical, Wilmington, MA, USA) featuring 8Tx-32Rx channels.

For the in vitro experiments, a human head-shaped phantom was fabricated using 3D printed resin³⁰. The phantom was initially filled with a phosphate-buffered saline (PBS) solution containing 10 mM L-glutamic acid, 10 mM creatine, and 20 mM D-glucose. To approximate the electrical and relaxation properties of human brain tissue, 4 g/L sodium chloride and 0.064 mM gadoterate meglumine (Dotarem) were added. To prevent bacterial degradation, 4 mM sodium azide was also included. The pH was adjusted to 7, and the phantom was heated overnight at 37 °C prior to the experiments.

Subsequently, the phantom was emptied and refilled with 3% porcine skin gelatin to enable APT-weighted CEST contrast, while the quantities of all other components remained unchanged relative to the PBS-based phantom.

For the in vivo experiments, four healthy volunteers (26 ± 5 years old, two males and two females) were scanned after providing written informed consent, in accordance with a French ethics committee.

2.2. CEST MRI methodology

2.2.1. Tailored pTx pulse design and implementation

Tailored pTx CEST saturation and excitation pulses were designed for each acquisition, for both in vitro and in vivo experiments.

To this end, a B_0 field map was acquired using a three-dimensional gradient-recalled echo (GRE) sequence, followed by B_1^+ field maps acquired with an interferometric turbo-FLASH sequence³¹, with 2.5 mm isotropic resolution and 5x5x2.5 mm resolutions with 2.5 mm slice gap, respectively (matrix size 128x128x72 and 40x64x40, respectively). B_0 shimming was performed at the beginning of each acquisition session using an in-house software based on quadratic programming¹⁴.

The CEST pulse optimization followed the methodology described by Delebarre et al.¹⁸ and consisted in designing pTx Gaussian saturation pulses composed of two complementary modes (A and B) to achieve effective saturation over a 1 cm axial slab positioned at the scanner isocenter. The two modes were applied in an alternating fashion and repeated N times until the desired total saturation time (T_{sat}) was reached. The optimization minimized the L_2 norm between the effective root mean square transmit field, $B_{1,\text{rms}}^+$, and a predefined B_1^+ target, under constraints of peak and average RF power and SAR (local and global). Two configurations were evaluated: i) protected mode (PM), in which conservative RF power limits are used to indirectly constrain SAR, and ii) VOP mode, in which RF energy limits and explicit SAR limits are applied. In the VOP mode, the average RF power per channel was limited to 3 W, compared to 1.5 W in PM. The total power limit (sum across channels) in the VOP mode was set to 18 W. In the protected mode, this limit was 8 W. In the VOP mode only, local SAR constraints were set to 20 W/kg and global SAR was limited to 3.2 W/kg, corresponding to the “Level 1” SAR mode of IEC³². All optimizations were performed in MATLAB (MathWorks, version 24.1, R2024a).

For readout of the CEST imaging sequence, dynamic RF shimming with three-spoke excitation²⁰ was employed for each slice.

The optimized saturation and excitation pulses were subsequently integrated into a two-dimensional multi-slice EPI CEST pulse sequence¹⁸, for which the CEST presaturation module was played once per repetition time (TR), right before the EPI slice selection.

For comparison purposes, pTx pulses were also optimized for 7 T in vitro acquisitions using the same power limits. The local SAR constraint was, however, set to 10 W/kg while the global SAR constraint remained 3.2 W/kg (“Normal” mode of IEC³²).

2.2.2. CEST acquisition parameters

For both in vitro and in vivo acquisitions, 5 slices (1.6 mm isotropic resolution, 0.3 mm slice gap) were acquired. The imaged field of view was 210 x 210 mm, with a matrix size of 128x128 and GRAPPA³³ acceleration factor 2. The EPI bandwidth was set to 1698 Hz/Px. Two sets of CEST pulses (Gaussians of duration 50 ms) were designed to provide efficient saturation of the target protons while maintaining frequency selectivity: “low- B_1^+ ” pulses for amide, guanidinium, and hydroxyl protons, and “high- B_1^+ ” pulses for glutamate protons.

For the low- B_1^+ acquisitions, 12 presaturation pulses (comprising 6 A-B pairs) were applied, resulting in an effective saturation time of 600 ms at a target $B_{1,rms}^+$ of 1.5 μ T. Sixty-six measurements were acquired corresponding to the following saturation frequencies: [-40, -40, -40, -10, -5.0:0.2:0, 40, 40, 40, 10, 5.0:-0.2:0] ppm. The frequency offsets ± 40 ppm were repeated three times to ensure steady-state conditions. Other imaging parameters were: TR = 2500 ms, echo time (TE) = 16 ms, flip angle (FA) = 73°. The acquisition duration was 2:53 min.

For the high- B_1^+ acquisitions, 6 presaturation pulses (comprising 3 A-B pairs), resulting in an effective saturation time of 300 ms at a target $B_{1,rms}^+$ of 2.5 μ T. Only for participant 3 (P3), the $B_{1,rms}^+$ target was 3.0 μ T and due to exceeded RF coil power limits, the acquisition volume for this contrast was reduced to 4 slices. Forty-six measurements were acquired with the following saturation frequencies: [-40, -40, -40, -10, -5.0, -4.0, -3.8, -3.5, -3.2:0.2:-1.8, -1.5, -1.0:0.2:0, 40, 40, 40, 10, 5.0, 4.0, 3.8, 3.5, 3.2:-0.2:1.8,

1.5, 1.0:-0.2:0] ppm. Due to the higher energy deposition of high- B_1^+ CEST saturation module, the TR was increased to 5000 ms, resulting in an acquisition duration of 4:05 min. Other imaging parameters were: TE = 16 ms, FA = 85°.

To correct for static field inhomogeneities, Water Saturation Shift Referencing³⁴ (WASSR) scans were acquired using either a high- B_1^+ or a low- B_1^+ pulse design, but with the pulse amplitude scaled down to reach $B_{1,rms}^+ = 0.3 \mu T$. Twenty-six measurements were acquired with saturation frequencies uniformly distributed between 1.0 and -1.0 ppm: [1.0, 1.0, 1.0:-0.1:0, -1.0, -1.0, -1.0:0.1:0]. Other imaging parameters were: TR = 1500 ms, TE = 16 ms, FA = 62°. The acquisition duration was 45 s.

To validate the performance of the CEST saturation module, a 1.5 μT low- B_1^+ pulse design was evaluated on the PBS-based phantom, with the pulse amplitude reduced to achieve five $B_{1,rms}^+$ targets: 1.5 μT , 1.4 μT , 1.2 μT , 1.0 μT and 0.5 μT . One hundred and six frequencies were acquired with the following saturation frequencies: [5.0, 5.0, 5.0:-0.1:0, -5.0, -5.0, -5.0:0.1:0]. The imaging parameters were the same as for the low- B_1^+ acquisitions.

In vivo, the relayed nuclear Overhauser effect (rNOE), arising from aliphatic protons from mobile proteins and macromolecules, was imaged with the low- B_1^+ pulse amplitude reduced by 33% to obtain $B_{1,rms}^+ = 1.0 \mu T$. The sixty-six frequency offsets and all other imaging parameters matched those for low- B_1^+ acquisitions.

For two participants (P2 and P4), an additional low- B_1^+ acquisition ($B_{1,rms}^+ = 1.5 \mu T$) with 16 presaturation pulses was added to study the effect of prolonged saturation ($T_{sat} = 800$ ms) on the CEST contrast. Other imaging parameters were: TR = 4500 ms, TE = 16 ms, FA = 73°. The repetition time had to be increased when increasing the saturation time to comply with SAR and system limits. The acquisition duration was 5:11 min.

Each in vivo CEST acquisition, with the exception of WASSR scans, was performed twice and later averaged.

To enable distortion correction using Topup^{35,36}, additional images were acquired during the vivo sessions with the same imaging parameters as the low- B_1^+ acquisition, but without the CEST saturation module. One image was acquired with the same phase-encoding direction as the CEST acquisitions (anterior-posterior) and a second image was acquired with the reversed phase encoding (posterior-anterior). A T_1 -weighted anatomical MP2RAGE³⁷ image with 0.6 mm isotropic resolution was also acquired on each volunteer for brain segmentation purposes, with the following parameters: TR = 5100 ms, inversion times = 991 ms / 2970 ms, bandwidth = 300 Hz/Px, FA = 7° / 3°, matrix size = 326 x 326.

2.3. Data processing

In vivo CEST acquisitions were motion corrected using FSL flirt³⁸. The first volume of each acquisition was aligned across acquisitions, after which all volumes within each session were realigned to the corresponding session's first volume. Both stages employed rigid-body transformation. From the two scans acquired with opposite phase encoding directions, a susceptibility-induced off-resonance field was estimated using FSL Topup^{35,36} and was applied to each CEST volume to correct for EPI distortions.

For both in vitro and in vivo acquisitions, voxel-wise frequency offset maps obtained from WASSR acquisitions were used to correct B_0 inhomogeneities by interpolating each CEST Z-spectrum to the corrected frequency offsets. Repeated in vivo CEST acquisitions were then averaged to improve the signal-to-noise ratio.

To generate the different CEST contrast maps, the magnetization transfer ratio asymmetry³⁹ (MTR_{asym}) at a given frequency offset $\Delta\omega$ was computed on a voxel-by-voxel basis:

$$MTR_{asym}(\Delta\omega) = \frac{S(-\Delta\omega) - S(\Delta\omega)}{S_{ref}} (1)$$

Where S is the acquired CEST signal and $S_{ref} = S(-40 \text{ ppm})$ is the unsaturated reference signal used for normalization.

For both in vitro and in vivo acquisitions, hydroxyl- and glutamate-weighted contrasts were generated by computing MTR_{asym} at three frequency offsets (-0.8, 1.0, and 1.2 ppm for hydroxyl protons, and 2.8, 3.0, and 3.2 ppm for glutamate protons) and averaging the resulting MTR_{asym} maps for each contrast. For each volunteer, APT-weighted, guanidinium-weighted and rNOE-enhanced maps were generated by computing MTR_{asym} at three discrete frequency offsets within defined ranges – 3.5 ± 0.1 ppm, 2.2 ± 0.2 ppm and -3.5 ± 0.5 ppm, respectively – and averaging the resulting maps.

Gray matter (GM) and white matter (WM) segmentations were performed on the T_1 -weighted images using sMRIprep⁴⁰ (version 0.18.0) and co-registered with the CEST images using custom Matlab scripts.

3. Results

3.1 Simulations

In vitro normalized circularly polarized (CP) B_1^+ profiles obtained at 7 T and 11.7 T are shown in Figures 1A and 1B, respectively. To evaluate the effectiveness of the pTx pulse design in mitigating B_1^+ inhomogeneities at 11.7 T and to compare it with that at 7 T, the normalized root mean square error (nRMSE) was computed for both the protected and the VOP modes as a function of $B_{1,rms}^+ \sqrt{DC}$, where $DC = T_{sat}/TR$ is the duty cycle (Figure 1D). For further comparison, the performance of the CP design under the same energy management strategies is also shown in Figure 1C. The results obtained for 7 T are consistent with previously reported findings^{18,23}. Across all pulse design strategies, 7 T optimization continuously yielded lower nRMSE values than at 11.7 T. This difference can be attributed to the more homogeneous initial CP B_1^+ profile at 7 T (Figure 1A) compared to that at 11.7 T (Figure 1B), which facilitates convergence toward lower nRMSE values during optimization. At 11.7 T, the use of VOP-based SAR constraints was essential to mitigate transmit field inhomogeneities, resulting in an average reduction of the pTx pulse nRMSE by 13% compared to the PM constraints across the total $B_{1,rms}^+ \sqrt{DC}$ range tested. For $B_{1,rms}^+ \sqrt{DC}$ values up to 1 μT , 11.7 T pTx pulses designed under VOP-based SAR constraints achieved similar performance to 7 T pulses, yielding average nRMSE values of 6.3% (vs 4.6% at 7 T).

Given the superior performances of VOP-SAR-constrained pulses, in vitro and in vivo acquisitions were only conducted with pulses optimized under these constraints.

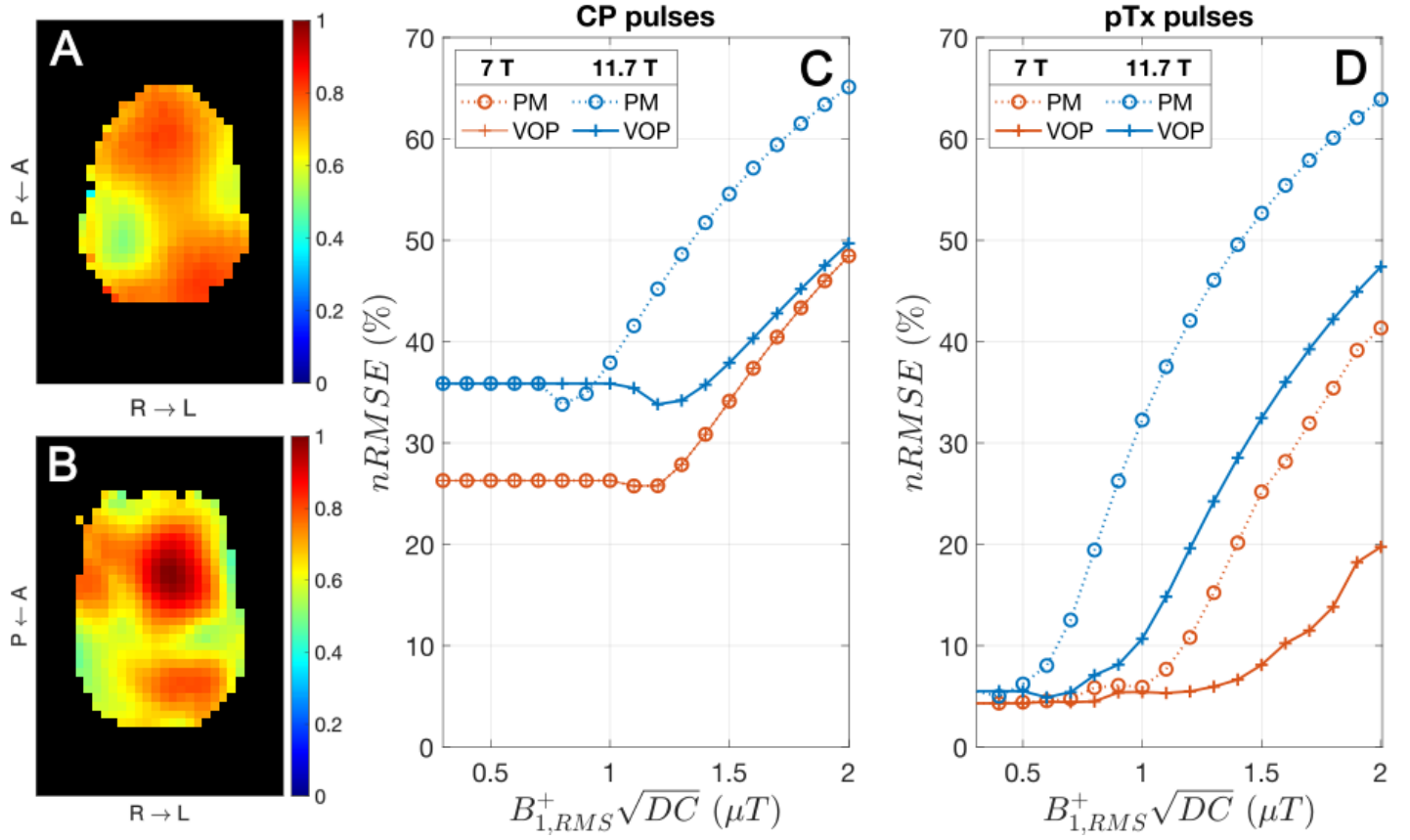


Figure 1: Normalized circularly polarized B_1^+ profiles at 7 T (A) and 11.7 T (B) on a gelatin-based human-head shaped phantom. Simulations of the normalized root mean square error (nRMSE) for circularly polarized (C) and tailored parallel transmission (pTx) pulses (D) at 7 T and 11.7 T, optimized under protected mode (PM) or virtual observation points (VOP) mode, are also shown.

3.2. In vitro

Experiments performed on the PBS-based phantom demonstrated clear spectral separation of glutamate, guanidinium, and hydroxyl protons across five B_1^+ amplitudes (Figure 2A). With the saturation parameters used, the resulting $B_{1,rms}^+ \sqrt{DC}$ values for low- B_1^+ and high- B_1^+ acquisitions were 0.73 and 0.61, respectively, corresponding to low nRMSE values as predicted by simulations (Figure 1D). For the gelatin-based phantom experiments, the pTx-VOP mode simulated B_1^+ profiles across the two slices

comprised in the 1 cm slab are shown for the low- B_1^+ (Figure 2B) and high- B_1^+ (Figure 2C) conditions, yielding average nRMSE values of 6.2% and 4.6%, respectively. CEST maps acquired on the gelatin-based phantom using these two pulse designs exhibited spatial inhomogeneity patterns consistent with the simulated profiles. However, the relative standard deviations measured for both the hydroxyl-weighted (Figure 2D) and glutamate-weighted (Figure 2E) acquisitions exceeded the predicted values, with mean MTR_{asym} of $4.3 \pm 0.57\%$ and $10.4 \pm 1.1\%$, corresponding to relative standard deviations of 10.8% and 13.3%, respectively. These discrepancies may be attributed to experimental factors not captured by the simulations, such as temperature gradients, density variations within the phantom, as well as additional contributions resulting from residual excitation inhomogeneity.

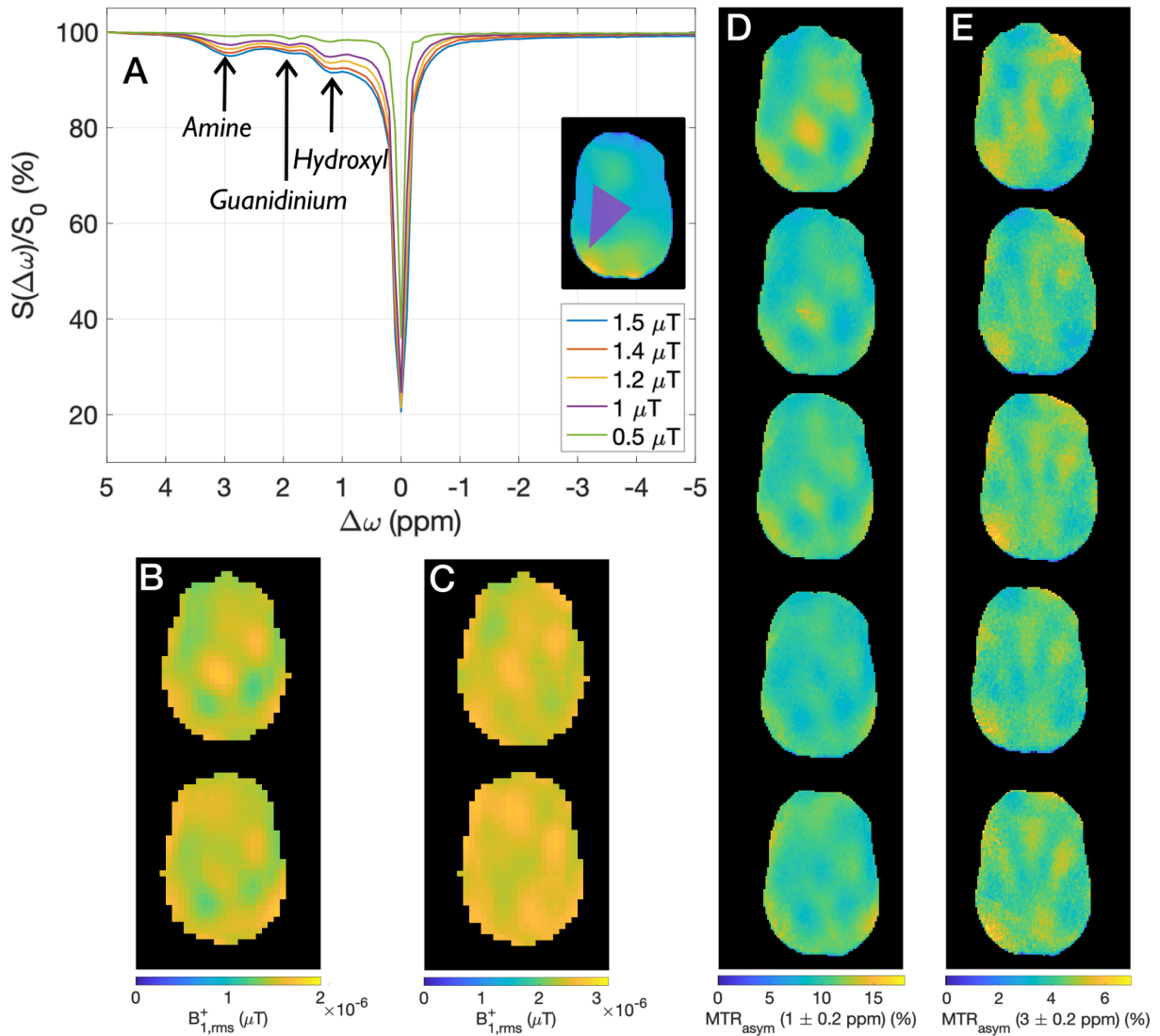


Figure 2: Results of in vitro acquisitions performed on the human 11.7 T scanner. Panel A shows Z-spectra averaged over a region of interest (purple triangle), acquired on the PBS-based phantom at different $B_{1,rms}^+$ amplitudes, showing clearly identifiable resonance peaks corresponding to glucose-hydroxyl, creatine-guanidinium, and glutamate-amine protons. Simulated $B_{1,rms}^+$ maps are shown in panels B (hydroxyl-weighted, $B_{1,rms}^+ = 1.5 \mu T$) and C (glutamate-weighted, $B_{1,rms}^+ = 2.5 \mu T$). The five-slice CEST maps acquired with optimized pTx pulses are shown in D (hydroxyl-weighted) and E (glutamate-weighted). The difference in slice number between the $B_{1,rms}^+$ and CEST maps results from the difference in slice thickness and slice gap (2.5 mm thickness with 2.5 mm gap vs 1.6 mm thickness and 0.3 mm gap).

3.3. In vivo

For the in vivo experiments, the optimized pTx-VOP mode CEST pulses yielded nRMSE values ranging from 4.3% (P4, low- B_1^+ pulses) to 12.1% (P2, low- B_1^+ pulses) (Table 1), indicating slightly reduced performance compared to phantom experiments. Nonetheless, MTR_{asym} maps demonstrated a clear contrast between GM and WM across all participants, in agreement with the T_1 -weighted images. MTR_{asym} maps of multiple CEST contrasts across all participants are shown in Figure 3 and all acquired CEST contrasts for P3 are shown in Supplementary Figure S1. CEST GM-WM contrasts were consistent across all participants, as reflected by the voxel-wise MTR_{asym} distributions in Figure 4.

Among the evaluated contrasts, rNOE-enhanced maps exhibited the strongest tissue differentiation, with a median GM-WM MTR_{asym} difference of $4.3 \pm 0.6\%$ across all participants. APT-weighted maps showed comparable median GM-WM contrast of $4.1 \pm 0.5\%$, with a median GM MTR_{asym} value of -5% . For low- B_1^+ acquisitions, the MTR_{asym} contrast in the 2–4 ppm range was strongly influenced by rNOE contributions, as illustrated by the mean Z-spectra in GM and WM regions of interest (ROIs) displayed in Figure 5A. Guanidinium-weighted CEST yielded a more moderate median GM-WM contrast of $2.3 \pm 0.2\%$ while hydroxyl-weighted CEST demonstrated minimal tissue contrast, with a median GM-WM difference of $0.7 \pm 0.2\%$ in MTR_{asym} .

Increasing the saturation duration to 800 ms enhanced both guanidinium- and hydroxyl-weighted GM–WM contrast by +0.3% in P2, accompanied by an increase of the median GM MTR_{asym} of +1.4% for both contrasts. The guanidinium-weighted MTR_{asym} maps for $T_{sat} = 600$ ms and $T_{sat} = 800$ ms are shown in Figures 6A and 6B, respectively. The distributions of GM MTR_{asym} values for guanidinium-weighted CEST in P2 at the two tested saturation times are shown in Figure 6C, demonstrating a shift toward higher MTR_{asym} values with increased saturation time, while the standard deviation remained unchanged. For P4, extending the saturation time resulted in only a modest gain in hydroxyl-weighted GM–WM contrast (+0.2%) and no measurable improvement for guanidinium-weighted CEST GM–WM contrast, but for both cases, the median GM MTR_{asym} increased by approximately 0.9%. Longer saturation times led to a slight decrease in APT-weighted GM–WM contrast for both participants (-0.7% for P2 and -0.1% for P4), primarily driven by moderate increases in the GM median MTR_{asym} values (+0.2% for P2 and +0.5% for P4).

Finally, glutamate-weighted CEST acquired at a saturation amplitude of 2.5 μT (Figure 3) yielded median GM–WM MTR_{asym} differences of 1.8% for P2 and 1.0% for P4. Mean Z-spectra over WM and GM ROIs are presented in Figure 5B. Increasing the B_1^+ amplitude further enhanced the contrast, reaching a GM–WM difference of 2.2% in P3 but also increasing the nRMSE to 12% (Table 1).

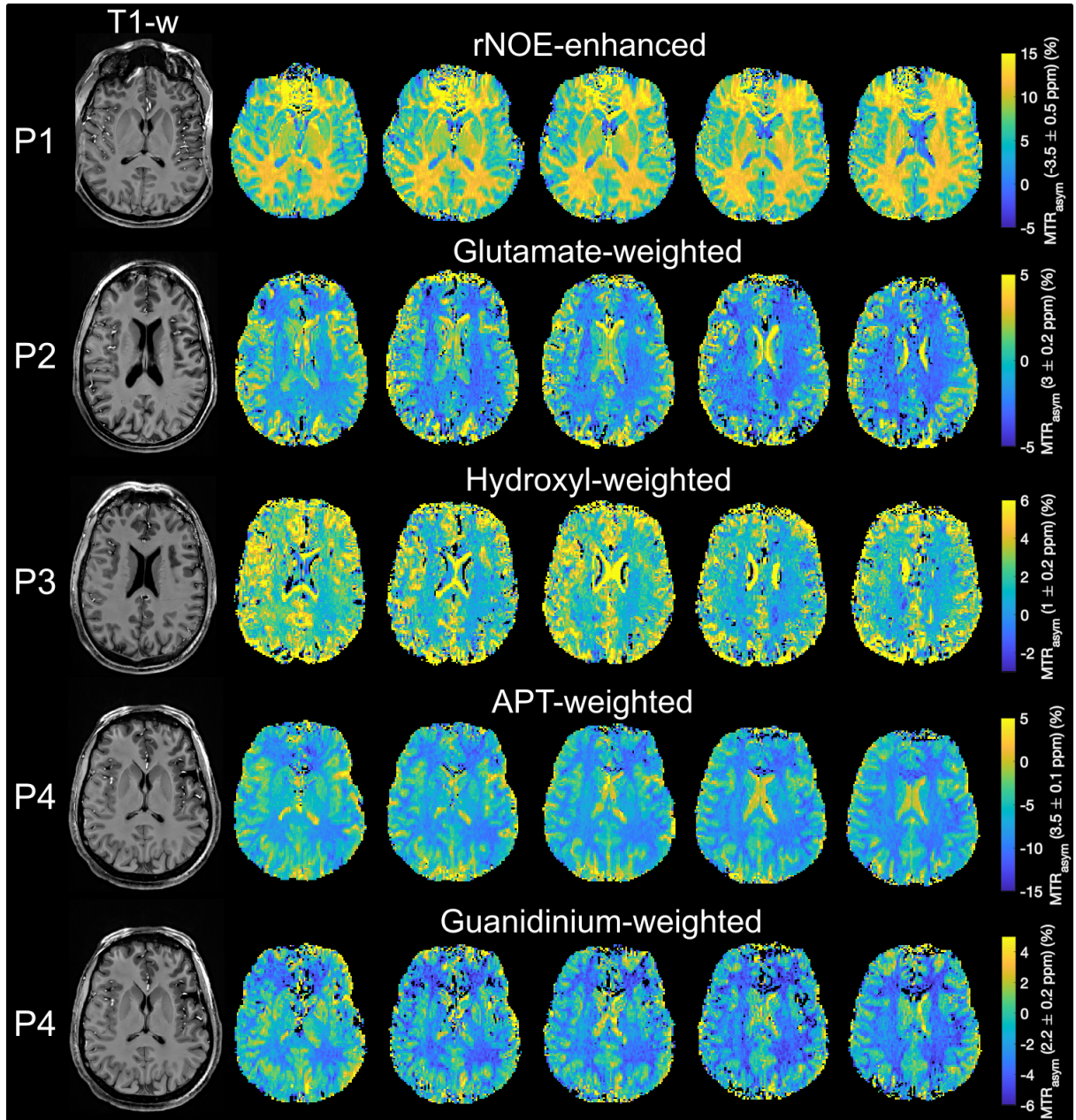


Figure 3: CEST MTR_{asym} maps for four different participants and different contrasts. Each row represents a different CEST contrast. From top to bottom: rNOE-enhanced CEST (participant 1), glutamate-weighted CEST (participant 2), hydroxyl-weighted CEST (participant 3), APT-weighted CEST (participant 4) and guanidinium-weighted CEST (participant 4). In the first column are shown the T₁-weighted anatomical images corresponding to the first slice of the acquired CEST volume.

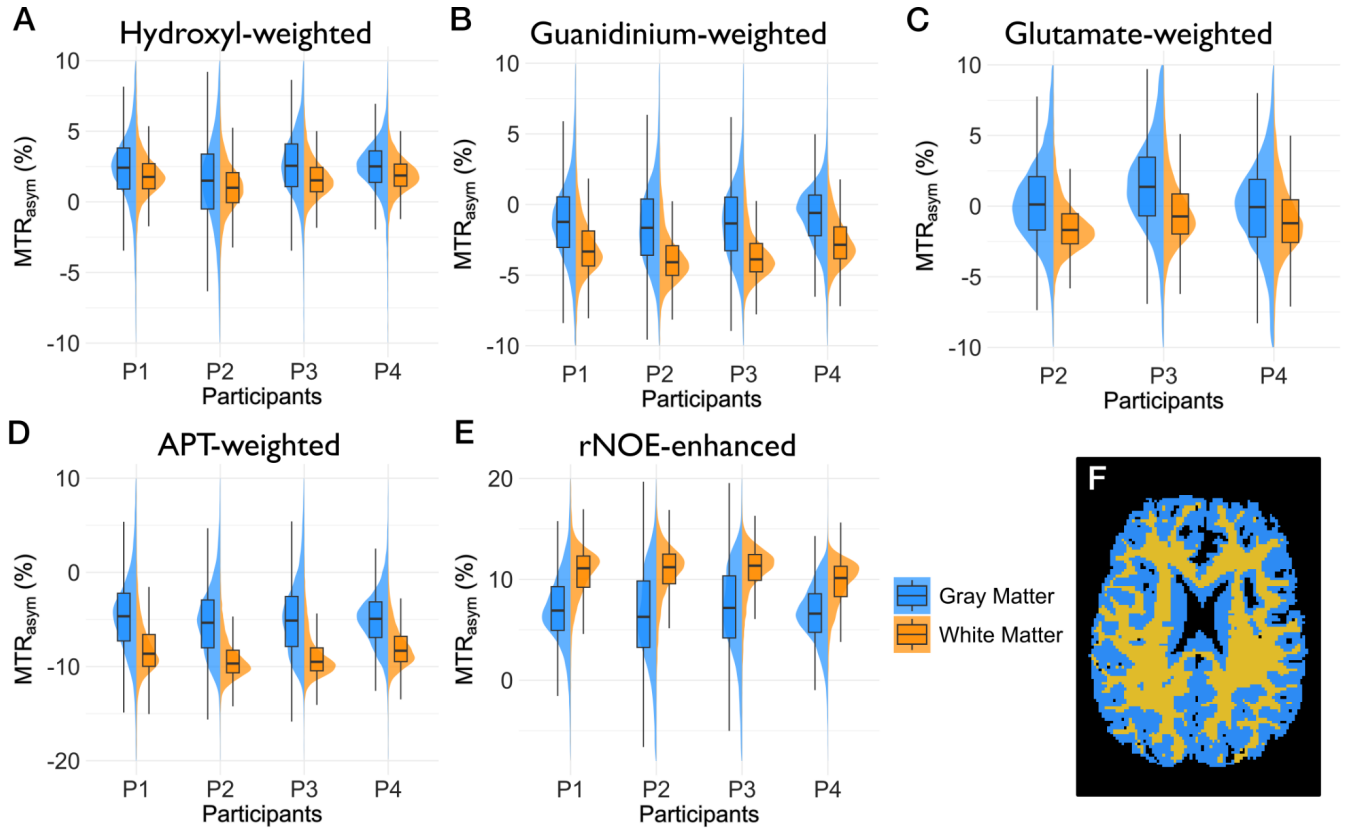


Figure 4: Voxel-wise distributions of gray matter (blue) and white matter (orange) MTR_{asym} values for the four participants for hydroxyl-weighted (A), guanidinium-weighted (B), glutamate-weighted (C), APT-weighted (D) and rNOE-enhanced CEST contrasts. An example of gray matter (blue) and white matter (orange) segmentation on participant 4 is shown in panel F.

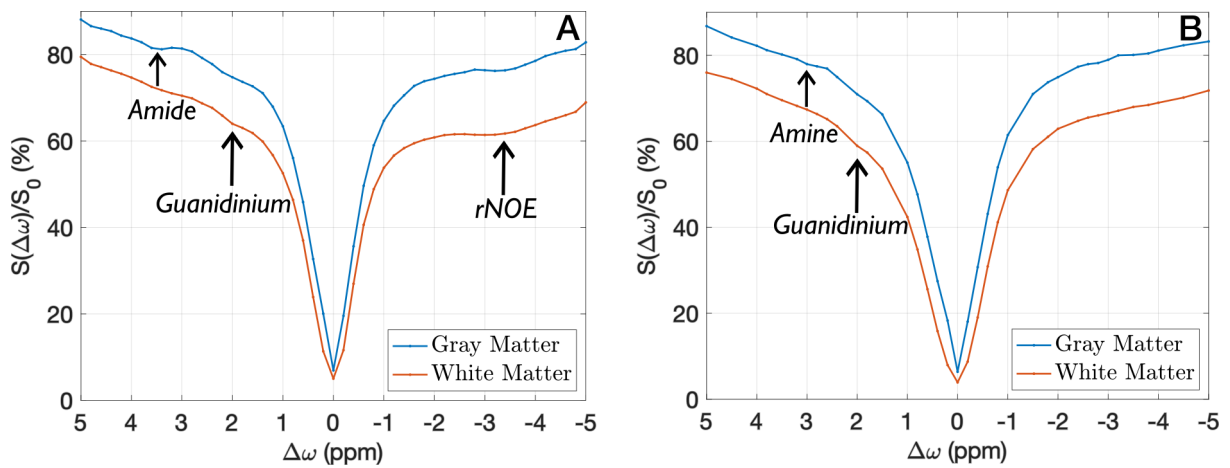


Figure 5: Z-spectra averaged on gray matter and white matter regions of interest on the low- B_1^+ acquisition of participant 2 (A). Panel B also displays gray matter and white matter Z-spectra but from the high- B_1^+ acquisition of participant 2.

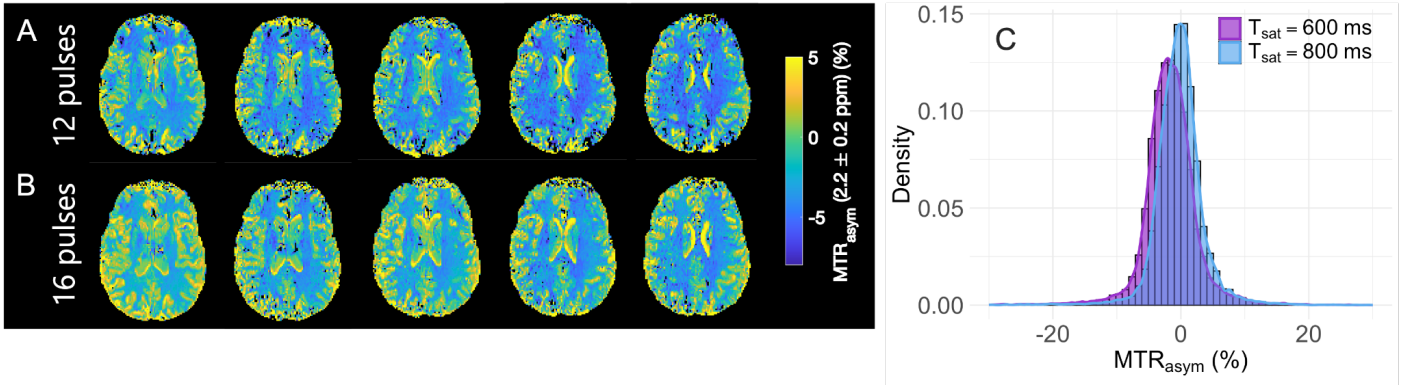


Figure 6: Effect of extended saturation time on guanidinium-weighted CEST MTR_{asym} maps in participant 2. CEST maps corresponding to T_{sat} of 600 ms (A) and 800 ms (B) are shown. For this contrast, the distributions of the MTR_{asym} in gray matter voxels indicate an increased CEST effect with longer T_{sat} (C).

A Z-spectrum-based two-point B_1^+ post-processing correction²⁸ was evaluated for the 1.5 μT datasets of each participant using acquisitions at 1.0 μT and 1.5 μT . Overall, this correction led to only marginal improvements in CEST contrast across participants and contrast weightings. The median MTR_{asym} values in GM and WM pre-correction and post-correction are listed in Supplementary Table S1 for all participants, for all evaluated contrasts. The largest post-correction differences were observed for P2, who also exhibited the highest pre-correction nRMSE (12.1%). Localized improvements in GM-WM contrast were especially seen in the left supramarginal and left superior parietal regions (Figures 7A-B, highlighted in red). The corresponding difference maps between images in Figure 7A and Figure 7B are shown in Figure 7C. Across the entire imaged volume for this participant, the guanidinium-weighted MTR_{asym} increased by 0.8% in WM and 0.5% in GM; however, the overall GM–WM MTR_{asym} contrast decreased by 0.4%. For all other participants, the differences were negligible. As an example, Figure 7D shows the difference between B_1^+ -corrected and uncorrected guanidinium-weighted MTR_{asym} maps in P1, demonstrating a highly homogeneous difference map with values close to zero.

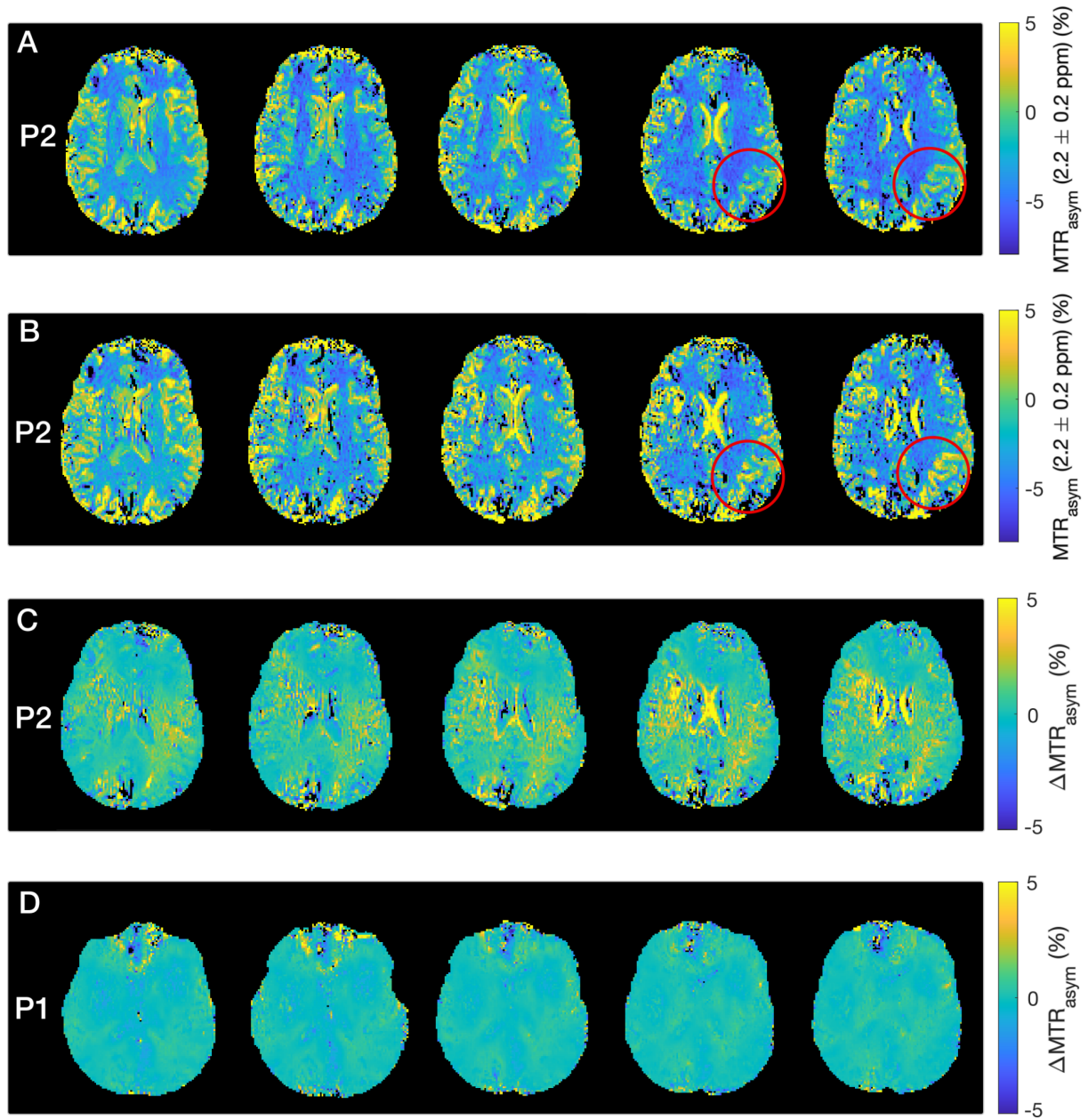


Figure 7: The post-processing B1⁺ correction did not yield measurable improvements in participants other than participant 2. For this participant, the corrected guanidinium-weighted CEST MTR_{asymp} maps (B) show localized gray-white matter contrast enhancement (red circles) compared with the uncorrected map (A). The MTR_{asymp} difference maps for participant 2 (C) exhibit larger correction-induced changes, whereas participant 1 shows minimal differences (D).

4. Discussion

This study reports the first in vivo human brain CEST images acquired at 11.7 T, demonstrating the feasibility of the method at this field strength. A tailored interleaved parallel transmission CEST pulse design under VOP-based SAR constraints was employed to mitigate transmit field inhomogeneities. Following initial validation on metabolite-doped PBS and gelatin phantoms, multiple contrast weightings were acquired at different saturation amplitudes in four healthy young volunteers, yielding measurable and consistent CEST effects across the cohort despite the increased technical challenges associated with ultra-high field MRI.

One of the main motivations for performing CEST MRI at high magnetic fields is the increased selectivity and detectability of metabolites¹⁷, driven primarily by the larger chemical shift dispersion between exchanging proton pools (in Hz). This increased dispersion reduces spectral overlap and spillover effects enabling a finer inspection of metabolite-related features in the Z-spectrum. In the present study, clear resonance peaks were observed for low- B_1^+ acquisitions, however no additional CEST contrasts beyond those typically reported at lower field strengths were detected.

The acquired CEST images exhibited clear gray-white matter differentiation across multiple CEST contrasts corresponding to different saturation offsets. The highest GM-WM contrasts were observed in APT-weighted and rNOE-enhanced images, with values comparable to previous studies at 7 T^{12,18,41,42}. Higher contrast for rNOE-enhanced images has been reported but without removing the magnetization transfer and direct saturation effects⁶. Guanidinium-weighted CEST imaging at 11.7 T revealed clear tissue differentiation across all participants, with GM-WM contrast exceeding that reported in previous studies^{25,43,44}. Glutamate-weighted GM-WM contrast was moderate but remained satisfactory, particularly at $B_{1,\text{rms}}^+ = 3 \mu\text{T}$, with values close to those recently reported at 7 T⁴⁵ with longer saturation times (800 ms). Human brain imaging of resonant metabolites around 1 ppm has been reported in only a few studies, and while hydroxyl-weighted images showed the weakest contrast at 11.7 T, the observed values still surpassed those previously reported^{18,46}.

Absolute CEST signal values were sometimes lower than those reported in prior studies, likely due to the shorter saturation times used in the present work. For example, APT-weighted CEST studies typically employ saturation durations between 1 and 3 s, and report correspondingly larger signal amplitude values^{12,47,48}.

Guanidinium-weighted CEST at 2 ppm has also been reported using a saturation time of 1500 ms, yielding higher CEST signal intensity values; however, these measurements were restricted to localized regions of interest and did not assess whole-brain GM–WM contrast⁴⁴. In the present study, increasing the saturation time from 600 to 800 ms yielded modest improvements with higher GM CEST signal intensity values in APT-, hydroxyl- and guanidinium-weighted CEST but no significant changes on the GM-WM contrast. For high- B_1^+ acquisitions, as previously suggested in the literature¹⁶, increasing the B_1^+ target to 3 μ T improved CEST signals and GM-WM contrast but this was achievable only with a reduced slice coverage due to the coil protection limits being exceeded and also resulted in increased saturation inhomogeneity.

These observations indicate that anatomical contrast is detectable under the present acquisition conditions at 11.7 T, highlighting the potential of ultra-high field CEST for targeted studies of localized brain regions in neurological diseases. At the same time, quantitative interpretation across studies remains complicated by the strong dependence of the CEST contrast on both tissue properties (pH, temperature, conductivity...) and acquisition-specific factors, including saturation scheme and field strength, which vary substantially across the literature^{49–51}.

In this work, the MTR_{asym} metric was selected as a simple and low-assumption approach for CEST contrast quantification. As previously discussed, this metric reflects overlapping contributions from multiple effects, including rNOE and magnetization transfer from macromolecules such as proteins and peptides. More advanced quantification metrics based on voxel-wise fittings such as Lorentzian difference analysis⁵², multi-pool Lorentzian fitting⁵³ or polynomial and Lorentzian line-shape fitting⁵⁴, have been proposed. While such methods may enable improved separation of individual CEST contributions, they are highly sensitive to the choice of initial fitting values and boundary conditions, which can lead to substantial variability in the resulting contrast and require careful interpretation⁵⁵. Moreover, for high- B_1^+ power, Lorentzian fitting approaches are not well-suited as the assumption of no interaction between the pools breaks down⁴⁹. The systematic evaluation of such additional post-processing strategies was beyond the scope of the present study, which was primarily

designed to establish the technical feasibility of CEST MRI on an 11.7 T system for humans and to acquire robust and reliable CEST images of the human brain.

As expected, the B_1^+ inhomogeneities were more pronounced at 11.7 T than at lower field strengths. Nonetheless, by allowing a higher local SAR (20 W/kg vs. 10 W/kg) the pTx-VOP mode enabled transmit field homogenization comparable to previously reported 7 T CEST optimizations^{18,23}, resulting in acceptable saturation homogeneity (typically smaller than 12.1% over a 1 cm slab). The retrospective B_1^+ correction provided minimal gains overall with the exception of very localized improvements observed in a single participant, consistent with previous observations by Völcke et al.²⁵. These results suggest that, in the context of effective pTx optimization, retrospective B_1^+ correction may offer limited additional benefit. This finding is particularly important for future high-field studies in patient cohorts, for which avoiding the need for additional datasets is critical to minimize total scan time.

In this study, saturation pulses were tailored on a per-subject basis to account for inter-individual B_1^+ variability; the development of universal pTx pulse designs applicable across subjects would substantially simplify clinical research deployment. Future improvements in pTx RF coil design may help achieve a more homogeneous B_1^+ field, which in turn may decrease the SAR and energy demand of CEST saturation and excitation pulses and allow more flexibility for the CEST saturation parameters.

5. Conclusion

This work demonstrates the feasibility of performing multi-contrast CEST MRI in the human brain at 11.7 T with 1.6 mm isotropic resolution using parallel transmission-optimized CEST saturation and excitation pulses. By combining custom sequence implementation with tailored pTx pulse design, we achieved robust and consistent GM-WM matter contrast across multiple CEST weightings in healthy volunteers despite the stringent RF and SAR constraints. Importantly, at 11.7 T, the combination of higher field strength and our optimized experimental conditions produced a clear increase in GM-WM contrast, which is likely a more meaningful measure of tissue-specific CEST effects than absolute signal values. Further advances in RF hardware, saturation pulse

design and optimization will improve robustness and spatial coverage, thereby fully realizing the potential of ultra-high field CEST MRI.

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Table 1: In vivo CEST pulse design performances

	P1 nRMSE (%)	P2 nRMSE (%)	P3 nRMSE (%)	P4 nRMSE (%)
Low- B_1^+ (1.5 μ T)	4.5	12.1	8.4	4.3
High- B_1^+ (2.5 μ T)		6.3		5.3
High- B_1^+ (3.0 μ T)			12	

Abbreviations: sRMSE, Normalized Root Mean Square Error; P, Participant

Figures and table captions:

- **Figure 1:** Normalized circularly polarized B_1^+ profiles at 7 T (A) and 11.7 T (B) on a gelatin-based human-head shaped phantom. Simulations of the normalized root mean square error (nRMSE) for circularly polarized (C) and tailored parallel transmission (pTx) pulses (D) at 7 T and 11.7 T, optimized under protected mode (PM) or virtual observation points (VOP) mode, are also shown.
- **Figure 2:** Results of in vitro acquisitions performed on the human 11.7 T scanner. Panel A shows Z-spectra averaged over a region of interest (purple triangle), acquired on the PBS-based phantom at different $B_{1,rms}^+$ amplitudes, showing clearly identifiable resonance peaks corresponding to glucose-hydroxyl, creatine-guanidinium, and glutamate-amine protons. Simulated $B_{1,rms}^+$ maps are shown in panels B (hydroxyl-weighted, $B_{1,rms}^+ = 1.5 \mu$ T) and C (glutamate-weighted, $B_{1,rms}^+ = 2.5 \mu$ T). The five-slice CEST maps acquired with optimized pTx pulses are shown in D (hydroxyl-weighted) and E (glutamate-weighted). The difference in slice number between the $B_{1,rms}^+$ and CEST maps results from the difference in slice thickness and slice gap (2.5 mm thickness with 2.5 mm gap vs 1.6 mm thickness and 0.3 mm gap).
- **Figure 3:** CEST MTR_{asym} maps for four different participants and different contrasts. Each row represents a different CEST contrast. From top to bottom: rNOE-enhanced CEST (participant 1), glutamate-weighted CEST (participant 2), hydroxyl-weighted CEST (participant 3), APT-weighted CEST (participant 4)

and guanidinium-weighted CEST (participant 4). In the first column are shown the T_1 -weighted anatomical images corresponding to the first slice of the acquired CEST volume.

- **Figure 4:** Voxel-wise distributions of gray matter (blue) and white matter (orange) MTR_{asym} values for the four participants for hydroxyl-weighted (A), guanidinium-weighted (B), glutamate-weighted (C), APT-weighted (D) and rNOE-enhanced CEST contrasts. An example of gray matter (blue) and white matter (orange) segmentation on participant 4 is shown in panel F.
- **Figure 5:** Z-spectra averaged on gray matter and white matter regions of interest on the low- B_1^+ acquisition of participant 2 (A). Panel B also displays gray matter and white matter Z-spectra but from the high- B_1^+ acquisition of participant 2.
- **Figure 6:** Effect of extended saturation time on guanidinium-weighted CEST MTR_{asym} maps in participant 2. CEST maps corresponding to T_{sat} of 600 ms (A) and 800 ms (B) are shown. For this contrast, the distributions of the MTR_{asym} in gray matter voxels indicate an increased CEST effect with longer T_{sat} (C).
- **Figure 7:** The post-processing B_1^+ correction did not yield measurable improvements in participants other than participant 2. For this participant, the corrected guanidinium-weighted CEST MTR_{asym} maps (B) show localized gray–white matter contrast enhancement (red circles) compared with the uncorrected map (A). The MTR_{asym} difference maps for participant 2 (C) exhibit larger correction-induced changes, whereas participant 1 shows minimal differences (D).
- **Table 1:** In vivo CEST pulse design performances

Supporting information captions:

- **Supporting Information Figure S1:** CEST MTR_{asym} maps acquired on participant 3: rNOE-enhanced (A), hydroxyl-weighted (B), APT-weighted (C), guanidinium-weighted (D), glutamate-weighted (E). The $B_{1,rms}^+$ target for the glutamate-weighted acquisition on this participant was 3 μT (compared to 2.5 μT for the other participants) and coil protection limits were reached; therefore, the slice volume was reduced to 4 slices instead of 5.
- **Supporting Information Table S1:** Median $\pm$ standard deviation of MTR_{asym}

values (%) in white matter (WM) and gray matter (GM) before and after B_1^+ correction, for all participants, for all tested contrasts.